

Edge Based Secure Smart Tags Generation Using PUF-ECC for Supply Chain Management with Hyperledger Fabric

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Abstract

Counterfeiting is a widespread issue of global magnitude, affecting almost every sector of the goods market, including fashion, food, pharmaceuticals, and digital media. The ongoing struggle against counterfeiting is a significant issue for corporations and commercial operations. Most of the existing supply chains rely on centralized authorities or intermediaries that lack the necessary security measures to ensure the prevention of counterfeiting and the capacity to track products. In order to address these concerns, this work aims to implement a supply chain based on blockchain technology that utilizes Physically Unclonable Function (PUF) devices integrated with NFC tags to provide traceability and prevent counterfeiting of products. The objective of this supply chain is to ensure the consistent identification and monitoring of products throughout the supply chain and to identify and detect any potential instances of counterfeiting. In addition, we integrated a blockchain-based supply chain that provides decentralization, high accessibility, and guarantees data integrity. In order to ascertain the legitimacy of the solution, two application scenarios have been implemented through chain codes on Resbarrypi connected with the NFC scanner module that show the solution's potential in low-power IOT devices, followed by separate components that constitute the solution.

Keywords

Physically Unclonable Function (PUF), Blockchain, Smart Tags, Chain Code, Anti-counterfeiting, Elliptic-Curve Cryptography (ECC), Decentralized Systems, Supply Chain Management

1. Introduction

The counterfeit goods sector poses a growing and substantial challenge, adversely affecting producers and consumers worldwide. The processes of globalization and the emergence of e-commerce have both contributed to the substantial intensification of this issue, leading to a decline in revenue for economic enterprises. Based on the 2020 compilation of the Counterfeit and Piracy Watch List by the European Commission, counterfeiting constitutes 3.3% of the whole turnover in the worldwide market [1]. Counterfeit components may pose significant security risks and lead to loss of human lives. This aspect becomes crucial when products are used in critical sectors such as the military, food, and medical industries [2, 3].

Indeed, counterfeiting has evolved from a predominantly physical phenomenon to a nearly unregulated realm. Consequently, it is crucial to establish agile and efficient approaches, designs, technologies, and laws to address this phenomenon. The prevailing belief among many is that counterfeiting is a "victimless" offense, implying that no one sustains any injuries during the act. Moreover, apart from the economic consequences, there are also grave ramifications linked to the expansion of counterfeit products on the market. Undoubtedly, certain counterfeit goods are intentionally supplied to customers who are aware of them and deliberately choose to buy inferior products that fail to fulfill the basic safety and security thresholds. Under other circumstances, consumers may be unaware they are acquiring counterfeit goods, potentially resulting in even more adverse consequences. It is possible to produce

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exact replicas of the original products through reverse engineering methods in both cases. These duplicates are then promoted as authentic to deceive the final purchaser and/or the regulatory bodies responsible for oversight [4].

This work proposes a supply chain system based on blockchain technology to tackle the traceability and anti-counterfeiting problems by using PUF-generated smart tags that are physically attached to products and stored on a private blockchain accessible to all consumers via NFC smart tags. Tags keys are stored on a Hyperledger Fabric blockchain that is accessible and immutable, assures product originality, and provides easy verification and tracking throughout the supply chain.

2. Background and Related Work

Blockchain is an innovative technology called Distributed Ledger Technology (DLT) that guarantees the transparency and immutability of transactions among several participants [5]. Despite the inevitable growth in size, it remains immutable, as written data cannot be modified or deleted unless the entire network becomes invalid. Numerous supply chain management solutions that utilize blockchain technology to address the aforementioned issues have been presented. These systems have been extensively examined in scholarly academic and industrial contexts.

A Physically Unclonable Function (PUF) is a hardware component that produces a unique and reproducible pattern of behavior for a device, determined by a particular input and set of conditions. This fingerprint functions as a highly distinctive identification and cannot be replicated [6]. A Physical Unclonable Function (PUF) depends on the intrinsic physical fluctuations that occur spontaneously during the fabrication of semiconductor devices. The inherent fluctuations, which are impossible to entirely eradicate regardless of the level of accuracy in the production process, result in arbitrary alterations to the operational parameters of the circuit. Instead of using a single cryptographic key, Physical Unclonable Functions (PUFs) employ a challenge-response authentication method to evaluate the physical microstructure. Upon introducing a physical input to the structure, the device's response is unpredictable due to the complex interaction between the input and the physical microstructure.

In the contemporary globalized society, the consumer supply chain has fundamentally transformed the modes of trade for commodities, enabling worldwide trading on an unparalleled magnitude. Blockchain technology has profoundly influenced the banking industry [7]. The ongoing development of this technology suggests its potential to address the enduring obstacles encountered by the consumer sector. The problems encompass monitoring products throughout manufacturing, addressing fraud and counterfeiting, and staying up-to-date with fast-changing consumer preferences. Nevertheless, the key challenges encountered by the consumer sector are ensuring safety, reducing the occurrence of counterfeit items, and maintaining cost-effectiveness [8].

This paper by Davor et al. [9] offers a comprehensive analysis of the blockchain idea and investigates its wide array of uses in logistics and supply networks.

In their groundbreaking study, Toyoda et al. [10] provide an innovative Product Ownership Management System (POMS) that utilizes RFID (Radio-Frequency Identification) tags to efficiently address the issue of counterfeiting in the post-supply chain. The authors propose an innovative protocol that enables seamless transfer and establishment of product ownership across several stakeholders, including the supply chain partners and purchasers, for objects fitted with RFID tags.

Experimental investigations were carried out by researchers on the proposed protocol through its implementation on the Ethereum platform. Within their work Naif Alzahrani et al. [11] introduce a fresh perspective on supply chain management, emphasizing a pioneering approach known as block supply. The present approach leverages cutting-edge developments in blockchain and Near-Field Communication (NFC) technology to establish a distributed supply chain. The authors' studies showcase the block-supply chain's extraordinary capacity to closely monitor items and identify various types of assaults, including modification, duplication, and tag reuse. Moreover, the authors have introduced a unique consensus technique that can efficiently handle expansive networks, ensuring scalability and security. The provided protocol exhibits remarkable efficiency and dependability when used in

networks of significant magnitude.

In their paper, Jinhua Ma et al. [12] present a method that employs blockchain technology to address the problem of counterfeit goods. The proposed strategy gives manufacturers the chance to provide genuine items without needing to manage their own retail outlets. Consequently, this might result in a significant reduction in costs related to guaranteeing perfect product quality. Nevertheless, techniques that depend on RFID tags may be vulnerable to cloning assaults, which might make this technology ineffective in deterring counterfeiting efforts [13, 14].

This work proposes a supply chain system based on blockchain technology to tackle the traceability and anti-counterfeiting problems using PUF (Physically Unclonable Function) and ECC (Elliptic-Curve Cryptography)-based devices. That tag is unclonable and secured on an immutable blockchain that provides complete originality and security.

3. Methodology

An edge-based authentication and tracking system deployed on Raspberry Pi has been developed. That scans each product and can generate bulk keys using an integrated NFC reader, generating a unique cryptographic hash using the PUF-ECC mechanism. The hash is a tamper-proof identifier for the product that we save along with metadata on the Hyperledger Fabric blockchain. Each product can be verified using hash, which enables real-time traceability and ensures product authenticity.

4. System Architecture

4.1. System Components

This architecture integrates low-power hardware with cryptographic primitives and a distributed ledger as follows.

4.1.1. Raspberry Pi 3 Model B+:

Used as an edge computing unit as well as a Hyperledger Fabric peer node.

4.1.2. NFC Reader (Connect via SPI Interface):

Connected to Raspberry Pi, it reads the unique identifier (UID) and metadata from NFC tags.

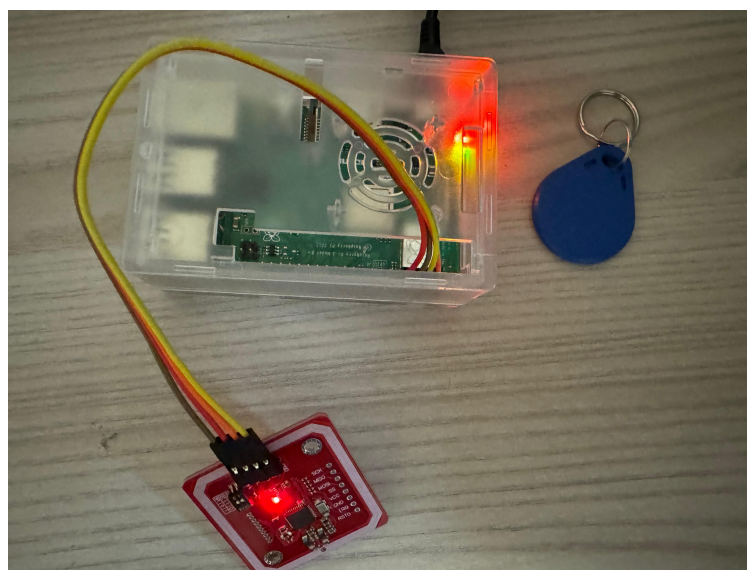


Figure 1: Our lab setup integration with Raspberry Pi and NFC Reader

4.1.3. PUF Module (Software via Pypuf):

To implement the challenge-response mechanism using arbiterpuf simulation to generate hardware-specific entropy.

4.1.4. ECC Key Generator (via `cryptography.hazmat.primitives.asymmetric.ec`):

Derives ECC key pairs deterministically from PUF response using SECP256R1 curves.

4.1.5. Hyperledger Fabric:

Permissioned Blockchain framework deployed locally on the Raspberry Pi to securely store and manage the smart tag records.

4.2. Smart Tag

The smart tag design, seen in Figure 2, consists of three primary components: (1) PUF, (2) ECC, and (3) NFC. The PUF component generates a digital fingerprint for the smart tag, which is expressed as a series of binary digits. The ECC component employs this fingerprint as a distinct cryptographic key to produce a public key and authenticate incoming communications. Furthermore, the NFC (Near Field Communication) component is the intermediary between the smart tag and the external environment. The proposed smart tag design leverages NFC technology integrated into mobile phones, enabling them to operate as RFID readers. This facilitates a product verification method at the customer level.

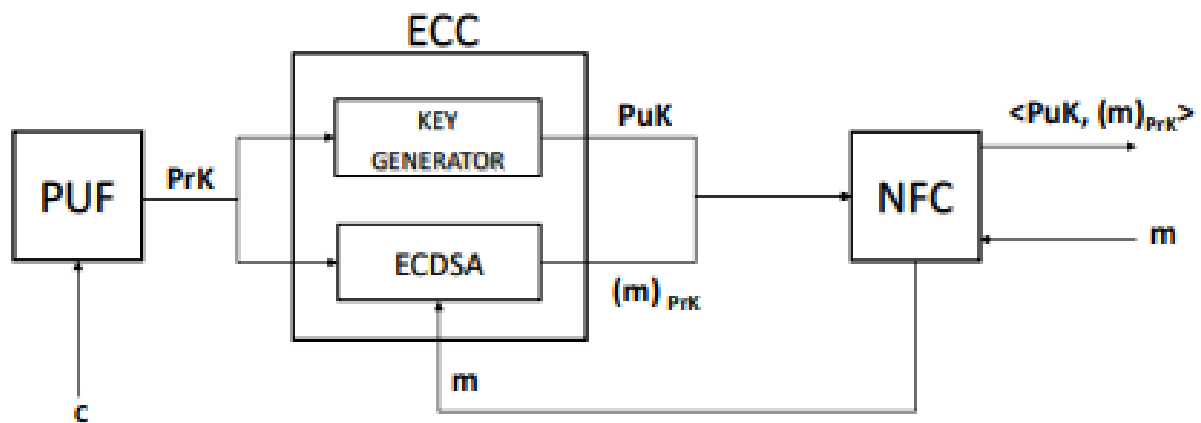


Figure 2: Smart tag Architecture

5. Potential Use Case Scenarios

The envisioned supply chain infrastructure, including blockchain technology and smart tags, may be customized to suit different application situations. Following, we examine two captivating scenarios:

- An uncoupled batch refers to a set of n items that are associated with n tags without any level of immediate integration between items and tags.
- coupled item refers to a luxury item associated with single tags, with immediate integration between items and tags.

Figure 3 shows the complete life cycle of the smart tags from the production line to the consumer verification process.

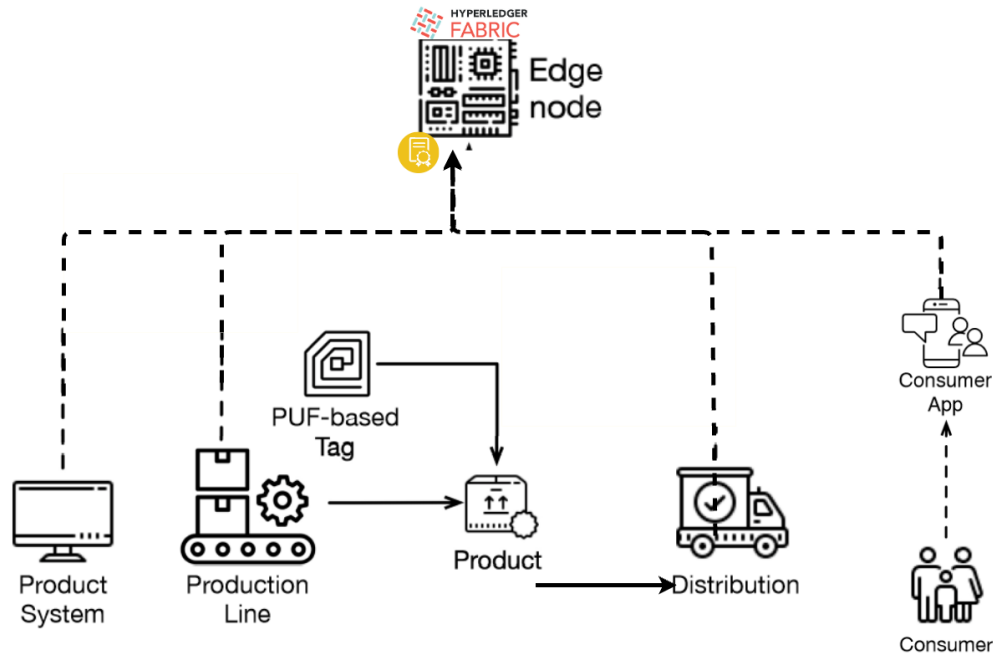


Figure 3: Suggested supply chain based on Blockchain technology aims to provide traceability and prevent counterfeiting of items using PUF

The smart tag application initially presents little logistical obstacles, yet a product can be replaced with a less desirable alternative. The implementation of trust mechanisms can alleviate this. The second scenario removes the need for product replacement by directly linking the product to its tag, which adds considerable complexity to the logistics, making it particularly suitable for luxury items.

6. Conclusion

The study presents a supply chain system that uses blockchain technology to monitor and mitigate the problem of counterfeit items. The proposed approach provides exceptional dependability and transparency against any risks to data integrity. To solve the aforementioned challenges, the proposed supply chain employs Physically Unclonable Function (PUF) and Elliptic-Curve Cryptography (ECC) based smart tags to establish unique identification and tracking of commodities across the supply chain. A comprehensive investigation of the solution's resilience in addressing specific threats has been conducted, and two distinct application scenarios have been established to assess its efficacy via smart contracts. As a possible direction for future study, the suggested method will be improved to tackle failure occurrences arising from the uncommon occurrence of creating private keys beyond the restricted range specified by the ECC algorithm. Furthermore, detailed analysis will be presented on evaluating complex harmful events, such as Denial-of-Service and Machine Learning assaults.

Declaration on Generative AI

During the preparation of this work, the author(s) used Chat-GPT-4 and Grammarly in order to: Grammar and spelling check the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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